Freight Rate Prediction

Machine Learning Assessment Report

1. Approach Overview

The objective of this assessment was to build a machine learning regression model to predict freight load rates (`posted_rate`) using the provided labeled development dataset.

The workflow included:

- Data loading
- Data cleaning and missing-value handling
- Feature engineering
- Train/test splitting
- Cross-validation
- XGBoost regression
- test evaluation
- Final predictions for the 12,000 validation records
- Fixed December 2025 predictions using the provided inputs

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2. Data Preparation

The development dataset contained **48,000 rows and 14 columns**, while the final validation dataset contained **12,000 rows and 13 columns**.

The target variable in the development dataset was:

`posted_rate`

The following data-cleaning steps were applied:

- Negative values in the `weight` column were considered invalid and converted to missing values.
- There were **292 negative weight values** in the training dataset and **145 negative weight values** in the validation dataset.
- Missing values in `weight` and `market_index` were handled using median imputation.
- The `date` column was converted to datetime format.
- A `month` feature was extracted from the date.

After cleaning, there were no remaining missing values in the processed datasets.

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3. Features

The model used the following numerical features:

- `pickup_lat`
- `pickup_lon`
- `delivery_lat`
- `delivery_lon`

- `distance`
- `weight`
- `market_index`
- `quote_signal`
- `month`

The categorical features were:

- `pickup`
- `delivery`
- `equipment`

The prediction target was:

- `posted_rate`

The `load_id` column was excluded from model training because it is an identifier rather than a predictive feature.

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4. Train/Test Split and Validation Strategy

To evaluate the model before generating the final validation predictions, the labeled development data was divided into training and holdout test sets.

An **80/20 train/test split** was used with:

```
random_state = 42
```

This resulted in:

- **38,400 training samples**
- **9,600 test samples**

The training set was used to fit the model, while the 9,600-row test set was kept as a holdout set for evaluating the model's performance.

In addition, **5-fold cross-validation** was used during model evaluation to provide a more reliable assessment across different subsets of the training data.

The main evaluation metrics were:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2

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5. Preprocessing

A preprocessing pipeline was used to apply the appropriate transformations to different feature types.

For numerical features, `StandardScaler` was used to standardize the values.

For categorical features, `OneHotEncoder` was used with:

```
handle_unknown="ignore"
```

The preprocessing steps and the regression model were combined into a single machine learning pipeline. This ensured that the same preprocessing operations were applied consistently during training and prediction.

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6. Model

The final model was **XGBoost Regressor**.

The main parameters were:

- `n_estimators = 200`
- `learning_rate = 0.05`
- `max_depth = 6`
- `objective = "reg:squarederror"`
- `random_state = 42`
- `n_jobs = -1`

XGBoost was selected as the final model for producing the validation predictions.

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7. Model Evaluation

The model was evaluated on the 9,600-row holdout test set.

The results were:

Metric	Result
MAE	**120.18**
RMSE	**548.98**
R^2	**0.8591**

The MAE of **120.18** means that the model's predictions were off by approximately $120 on average on the holdout test set.

The R^2 score of **0.8591** indicates that the model explained approximately **85.9% of the variance** in the target variable on the holdout test set.

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8. Final Validation Predictions

After evaluating the model, it was used to generate predictions for all **12,000 rows** in the provided validation dataset.

The generated file contained:

- `load_id`
- `predicted_rate`

All **12,000 predictions were successfully generated**, with no missing predicted values.

The provided Spotter scoring script confirmed:

Validated 12,000 final predictions.

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9. Fixed December 2025 Predictions

The provided December input dataset was used to generate the required fixed December prediction chart.

The fixed shipment scenario represents:

- Route: **Lexington → Fort Wayne**
- Distance: **360 miles**
- Equipment: **Dry Van**
- Weight: **32,000 lb**
- Prediction period: **December 2025**

The model generated predictions for all **31 days of December 2025**.

The provided scoring script confirmed:

Validated 31 fixed December predictions.

The scorer also successfully created:

`scorer_results/candidate_december.png`

December Prediction Chart

[INSERT `candidate_december.png` HERE]

Figure 1. Candidate December 2025 Predicted Load Rate

The chart shows a nearly constant predicted load rate across the December dates. This behavior is expected because the fixed December inputs keep the main shipment characteristics unchanged, while the model does not use the individual day of the month as a separate predictive feature.

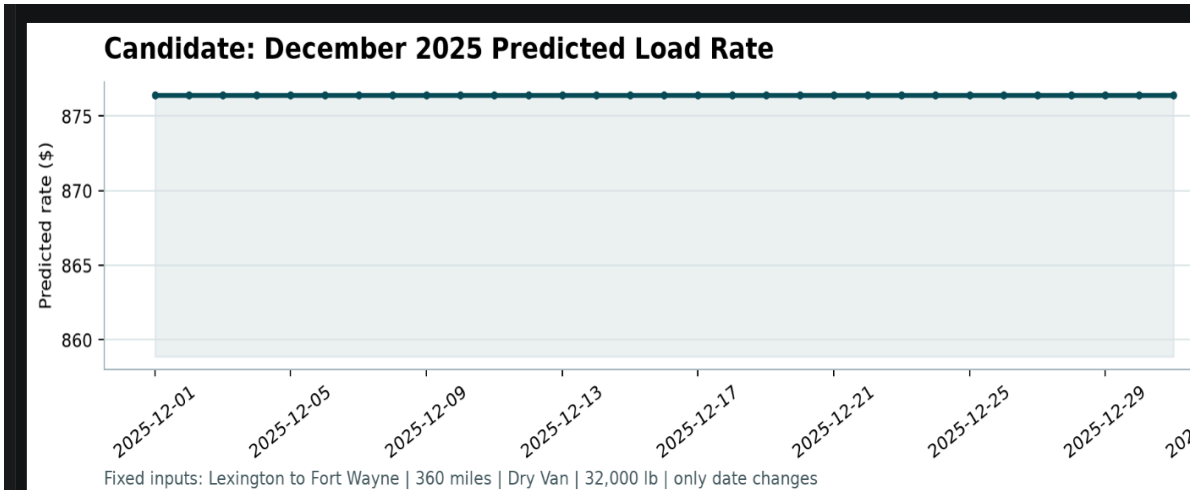
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10. Scoring Validation

The provided `score.py` script completed successfully.

The final output was:

- **Validated 12,000 final predictions.**
- **Validated 31 fixed December predictions.**
- **Created chart: `scorer_results\candidate_december.png`**



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11. Conclusion

The final machine learning workflow included data cleaning, feature engineering, an 80/20 train/test split, five-fold cross-validation, preprocessing of numerical and categorical variables, and XGBoost regression.

The final model achieved:

- **MAE: 120.18**
- **RMSE: 548.98**
- **R^2: 0.8591**

The model successfully generated predictions for all **12,000 validation records**, and the provided scorer validated all **31 December predictions** and generated the required December prediction chart.

The final Spotter validation metrics for the submitted predictions will be calculated by Spotter after submission.